

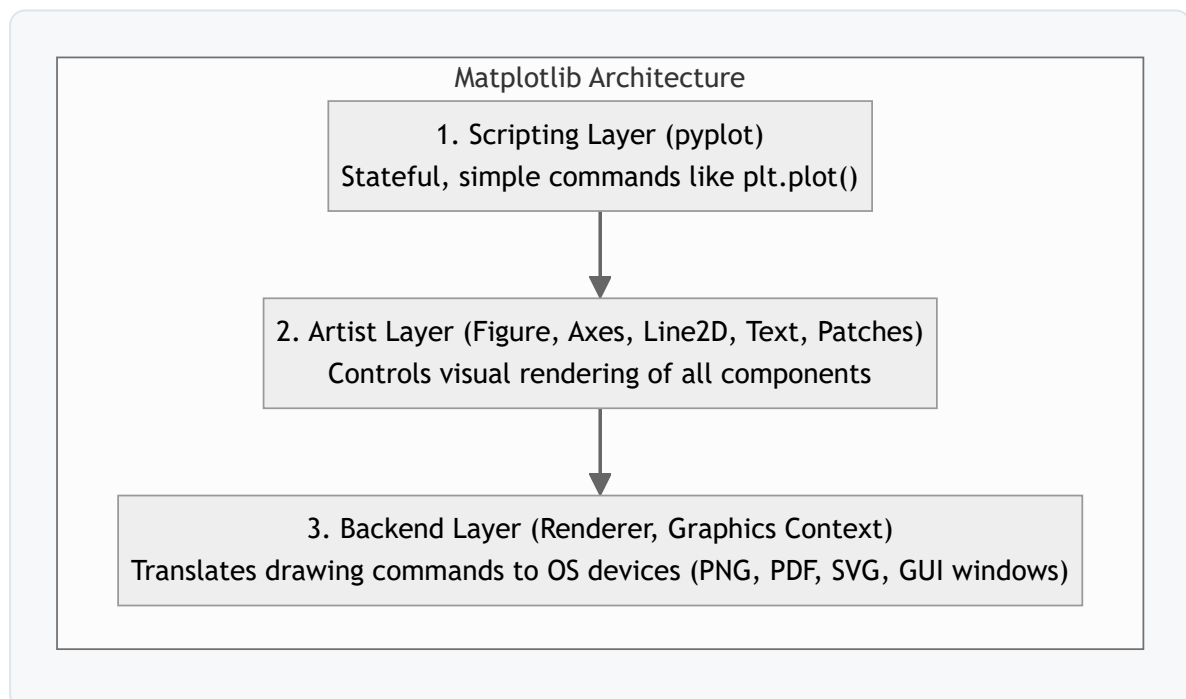
Day 18 - Lecture 18.3: Introduction to Matplotlib

1. History & Architecture

Matplotlib is the bedrock visualization library in the Python scientific stack.

- **Created by:** John D. Hunter in 2003.
- **Motivation:** Designed originally to visualize electroencephalography (EEG) signals with a syntax familiar to **MATLAB** users.
- **Ecosystem:** Foundation upon which Pandas plotting, Seaborn, Yellowbrick, and many geospatial libraries are built.

2. The 3-Layer Architecture of Matplotlib



- 1. Backend Layer:** Communicates with the hardware/operating system window (e.g., Agg for PNG raster images, PDF/SVG for vector graphics, TkAgg/Qt for GUI interaction).

2. **Artist Layer:** Everything seen on the screen is an **Artist** (lines, rectangles, text, spines, tick marks).
 3. **Scripting Layer (`matplotlib.pyplot`):** A collection of functions that make Matplotlib work like MATLAB, maintaining state across calls.
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3. Installation & Standard Import Convention

```
pip install matplotlib
```

Standard Python import convention:

```
import matplotlib.pyplot as plt  
import numpy as np
```

4. Key Takeaways

- `pyplot` is a module inside `matplotlib` providing the stateful scripting interface.
- Always use `import matplotlib.pyplot as plt`.